# Weekly Student Progress Performa

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| **Team Id** | CS26084 | **Week no / Date** | Week 8 07 September 2026 |
| **Project Title** | SentinelCore: A Network Detection and Incident Response Platform | | |
| **Members Present** | Om Rashiya (24cs084), Yash Patel (24cs082) | | |
| **Mentor** | Akshita Kadam | | |

## WEEK 8 • Faceted Search and Correlation Rules

| Section | Details to be filled by the team |
|---|---|
| **Planned Tasks (from last week)** | Add scalable event exploration and a correlation engine that can turn raw alerts into actionable findings. |
| **Tasks Completed This Week** | Designed keyset-paginated event queries, filter/facet contracts, saved searches, and correlation rules stored as data. Implemented evaluator patterns for threshold, sequence, rare-event, and beacon-like behavior, plus candidate creation and rule-run tracking. |
| **Commits Made** | Total: 0 standalone commits this review period. Search and correlation were integrated as feature milestones and committed after joint testing on 21 September. |
| **Issues Created / Closed** | Created: Offset pagination degrades on large event tables; hard-coded detectors are difficult to tune. Closed: Adopted indexed keyset pagination and editable rules-as-data with bounded evaluator inputs. |
| **Milestone Status** | ~63% complete — on track. Analysts could search evidence and define repeatable detections. |
| **Blockers / Challenges Faced** | Correlation needed deterministic windows and duplicate suppression so repeated worker passes would not create duplicate candidates. |
| **Plan for Next Week** | Complete incident promotion, assignment, status/severity changes, evidence linking, and immutable history. |

# Research and Technical Study

**Research objective.** Add scalable event exploration and a correlation engine that can turn raw alerts into actionable findings.

**Study completed.** Studied keyset versus offset pagination, time-window correlation, threshold and sequence detection, beacon periodicity, and false-positive control. Rule state and runs are persisted so detections remain explainable and auditable.

## Correlation evaluator comparison

| Evaluator | Pattern represented | Explainable output |
|---|---|---|
| Threshold | Count within a window | Group, count, and matched events |
| Sequence | Ordered event stages | Completed steps and evidence |
| Rare | Uncommon grouped activity | Observed group and baseline window |
| Beacon | Repeated periodic activity | Intervals, tolerance, and events |
| All types | Repeated worker passes | Rule/run records and duplicate suppression |

## Decision carried into the project

Designed keyset-paginated event queries, filter/facet contracts, saved searches, and correlation rules stored as data. Implemented evaluator patterns for threshold, sequence, rare-event, and beacon-like behavior, plus candidate creation and rule-run tracking.

# Implementation Record and Website Evidence

## Search contract

Defined bounded time windows, severity/type/address/port/protocol/signature/text filters, opaque cursors, result limits, sort direction, cached facets, and reusable saved searches.

## Rules as data

Stored rule name, description, enable state, type, match filters, grouping fields, window, threshold, severity, duplicate-suppression window, and type-specific parameters in persistent records.

## Evaluator types

Implemented threshold, sequence, rare-event, and beacon-style evaluation. Candidates retain the rule and matched event evidence so an analyst can understand why the detection was produced.

## Correlation workspace

Built Rules and Candidates tabs, a New rule action, list columns for type/window/threshold/candidate count/last run, and an empty state that directs administrators to create the first rule.

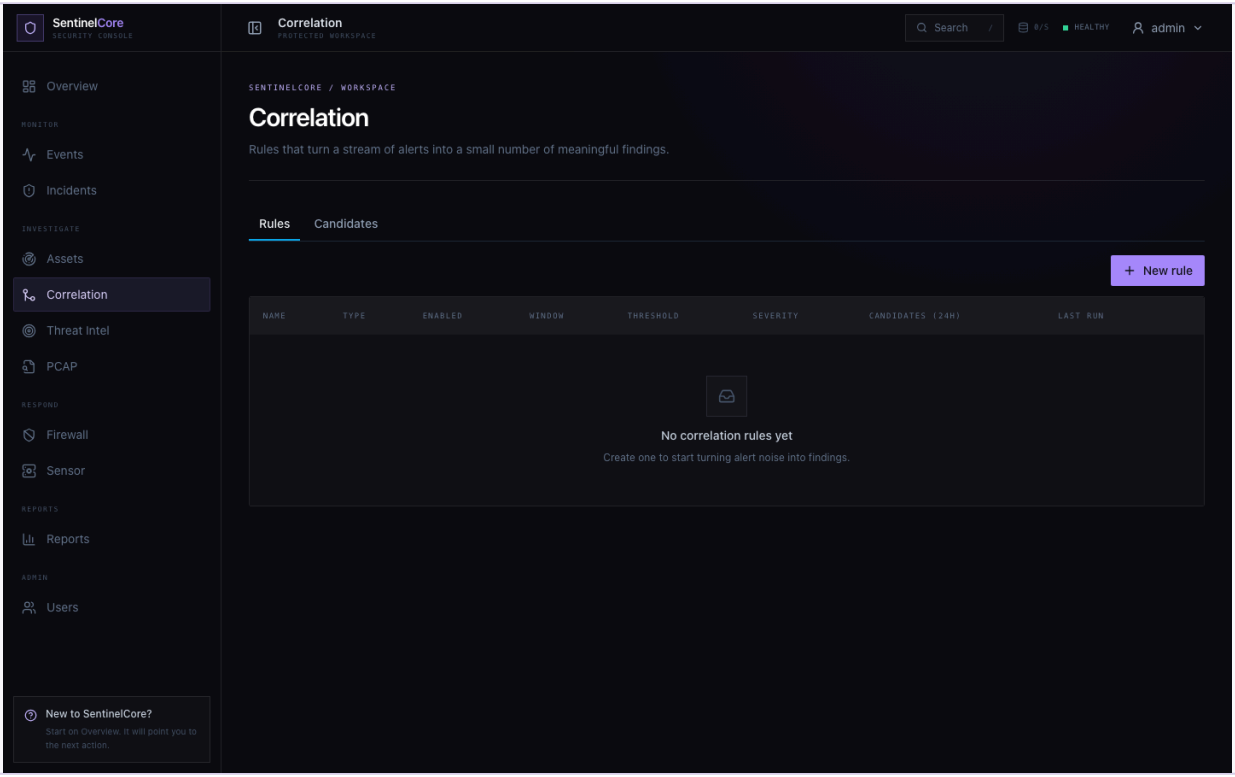



*Figure W8.1 — Project-owned SentinelCore interface mapped specifically to Week 8.*

**Evidence interpretation.** The Correlation workspace shows the Rules/Candidates organization, rule creation entry point, operational columns, and the explanatory empty state. Its visible controls, labels, status fields, and navigation provide direct evidence for the work described on this page.

# Repository and Architecture Evidence

**Correlation workspace implementation**

frontend/src/pages/correlation/CorrelationPage.jsx

```
  { key: "candidates", label: "Candidates" },
];

export default function CorrelationPage() {
  const [tab, setTab] = useState("rules");

  return (
    <>
      <PageHeader
        title="Correlation"
        description="Rules that turn a stream of alerts into a small number of meaningful findings."
      />

      <div className="mb-4 flex gap-1 border-b border-slate-800">
        {TABS.map((t) => (
          <button
            key={t.key}
            onClick={() => setTab(t.key)}
            className={`border-b-2 px-3 py-2 text-sm font-medium transition-colors ${
              tab === t.key
                ? "border-sky-500 text-slate-100"
                : "border-transparent text-slate-400 hover:text-slate-200"
            }`}
          >
```

Repository evidence captured directly from the SentinelCore workspace

*Figure W8.2 — Correlation workspace implementation. This is a separate internal artifact, not a copy of the website image.*

## Repository files used for traceability

- reports/modules/M06-search_1.md
- reports/modules/M07-correlation_1.md
- frontend/src/pages/correlation/CorrelationPage.jsx
- backend/app/api/routes/correlation.py

## Verification

Rule types, fields, candidate behaviour, and user-facing labels were verified against the M6/M7 specifications and implemented backend/frontend files.

**Traceability conclusion.** Correlation workspace implementation directly supports the Faceted Search and Correlation Rules milestone. Rule types, fields, candidate behaviour, and user-facing labels were verified against the M6/M7 specifications and implemented backend/frontend files. The artifact, source paths, implementation record, and website evidence therefore describe the same weekly scope.

# Weekly Closure and Faculty Assessment

## Completed outcome

Designed keyset-paginated event queries, filter/facet contracts, saved searches, and correlation rules stored as data. Implemented evaluator patterns for threshold, sequence, rare-event, and beacon-like behavior, plus candidate creation and rule-run tracking.

## Commit record

Total: 0 standalone commits this review period. Search and correlation were integrated as feature milestones and committed after joint testing on 21 September.

## Issue and resolution

Created: Offset pagination degrades on large event tables; hard-coded detectors are difficult to tune. Closed: Adopted indexed keyset pagination and editable rules-as-data with bounded evaluator inputs.

## Milestone position

~63% complete — on track. Analysts could search evidence and define repeatable detections.

## Challenge handled

Correlation needed deterministic windows and duplicate suppression so repeated worker passes would not create duplicate candidates.

## Next-week hand-off

Complete incident promotion, assignment, status/severity changes, evidence linking, and immutable history.

## Repository reference

https://github.com/rashiyaom/SentinelCore

| Assessed By (Faculty) | Marks Awarded (5 marks) | Faculty Remarks |
|---|---|---|
| | | |